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Coffee Management Database

Database Project Report

Agricultural Services Database for Coffee Farmers Support

Group	Group 4
Faculty	Engineering, Design and Technology
Department	Computing & Technology
Course	Database Programming
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CHAPTER 1: PROJECT DESCRIPTION

1.1 Project Background

Coffee in Uganda is one of the major agricultural products contributing economically to the country with millions of shillings generated from its production activities in various regions. Despite its significant contribution, the agricultural sector is still experiencing problems such as inadequate record-keeping, irregular distribution of inputs, and absence of a well-structured communication system between farmers and service providers. The Ministry of Agriculture has on several occasions advised the use of digital tools to nip these inefficiencies in the bud such as farmer registration, tracking of input distribution, and monitoring of production volumes at plot level. Our project is a direct answer to that need. Group 4 has created and implemented a completely normalized relational database system the Coffee Management Database which integrates farmer profiling, farm and plot management, production tracking, training programmes, agricultural input distribution, and extension worker management into one platform.

1.2 Problem Statement

Core Problem: The Ministry of Agriculture does not have a unified system for registering coffee growers, tracking production, distributing farm inputs, and linking farmers with extension workers. This project will develop and build a secure Agricultural Services Database to solve these problems and facilitate data-driven decision-making in Uganda's coffee sector.

1.3 Project Objectives

The main aim of this project is to create a database system that will work efficiently for the day-to-day operations of coffee farming service programs. In particular, the system is intended to:

1. Establish and maintain a unified and reliable source of farmer information by using National Identification Number (NIN) as the main reference.

2. Allow for detailed logging of farm information, such as GPS location, size of the area, and detailed breakdowns of different parts of the farm.

3. Record and maintain production data on a per-plot basis, noting the harvest season, amount (in kilograms), and quality level.

4. Keep track of and verify the supply of farming inputs and seedlings to registered farmers.

5. Through database-level triggers, maintain data consistency, and through stored procedures, automate regular work processes.

6. Develop a robust and safe role-based access control system where each user group obtains only the level of information necessary for performing their duties.

01 MILESTONE ONE

Requirements Gathering & Analysis

2.1 Functional Requirements

The following functional requirements define what the system must be able to do at an operational level.

2.1.1 Farmer and Farm Profiling

- The system stores the GPS coordinates and the total area of each farm to allow for geographic mapping of production areas.
- One farmer might own and manage several farms; each farm is accounted for separately, having its own records and plots.
- The unique National Identification Number (NIN) of each farmer is used as the primary key in the PERSON entity.

2.1.2 Production Tracking

The system will log coffee harvest season data per farm, such as the kind of coffee (Robusta or Arabica), quantity in kilograms, and quality level. Through view-based aggregation, it is possible to calculate the yield trends year over year not only for individual farmers but also for the geographical regions where their farms are located. A trigger is in place to stop harvest data from being recorded for plots that do not exist in the system, thus maintaining referential integrity.

2.1.3 Input and Seedling Distribution

The system monitors the stock levels of both seedlings and fertilizers live. Farmers cannot request more seedlings than the size of their registered farmland limit, as the system enforces this rule to allow a fair distribution of the inputs. A detailed log is maintained for each handing over including the farmer receiving, the input type given, amount given, handing over location, and the date.

2.1.4 Extension Service Management

Extension officers are given particular areas (sub-counties or parishes) to work in. Each piece of advice is recorded with the date, the farmer met, and the exact recommendation given (for example, Pest Control, Irrigation). Staff can only record their field trips in zones where they are assigned.

2.1.5 Reporting and Analytics

The system creates monthly Support Distribution reports identifying the farmers who have been given inputs. Ministry can plan export volumes using Yield Forecast reports which are based on plot-level production data. Three database views allow abstraction over complex multi-table joins for common reporting queries.

2.1.6 Security and Access Control

Different database users system accounts are supported by the system, one for each operational role. In line with the Principle of Least Privilege, each role is given only the minimum privileges required for its job to ensure security. An audit log automatically registers changes that are made to the data, including who, what, and when, for all sensitive tables. Complex multi-table operations are totally hidden from users by means of stored procedures, so that data manipulation is done only through authorized points of entry.

2.2 Data Requirements

The system manages the following core data entity groups:

Data Category	Key Data Captured
Personal & Farmer Information	Full name, gender, date of birth, phone, email. Farmer-specific: land size, sub-county location, cooperative membership.
Staff & Extension Services	Extension worker qualifications, roles (field officer or trainer), assigned geographic zone, linked visits and training sessions.
Farm & Production Data	Farm GPS coordinates, total area. Plot-level: coffee variety, area (acres), planting year, tree count. Production: quantity (kg) and quality grade per harvest season.
Agricultural Inputs & Suppliers	Input types (fertilizers, seedlings), source quantities, unit costs. Supplier name, location, and certification status.
Training & Farmer Participation	Programmes topics, duration, venue. Farmer enrolment records with completion status.
Farm Visits & Reporting	Visit date, assigned officer, field observations, and recommendations captured in a linked report.
Geographic & Administrative Data	Hierarchical structure: Region → District → Sub-county. Pre-loaded to support farmer and farm location mapping.
Cooperatives & Distribution	Cooperative membership and distribution event records for resource allocation and coordination tracking.
System Monitoring & Audit	Audit log entries: action type, affected table, old and new values, changed-by user, and timestamp.

2.3 Assumptions

The following assumptions were made to bound the scope and enforce consistency across the system:

1. A farmer has to be an adult minimum of 18 years. This limitation is guaranteed on the database through a BEFORE INSERT trigger on the PERSON table.
2. At the time of registration, a National Identification Number is considered valid. The system is not linked to NIRA for real-time NIN verification.
3. The geographical hierarchy (Region District Sub-county) is a reflection of Uganda's administrative structure and is loaded into the system before use.
4. A field officer and a trainer are the two roles that an extension worker may take, but he cannot be both at the same time (disjoint specialisation).
5. A person may be a farmer and an extension worker at the same time since the PERSON table is a shared supertype linked by separate foreign keys.
6. No change in the coffee variety within a plot from planting to harvesting. Changes of variety imply a new plot record.
7. All prices are in Uganda Shillings (UGX).
8. Only suppliers who have been approved by the Ministry may bring goods into the system.
9. It is necessary to create a farm visit record before any linked distribution event can be processed.
10. Ugandan Coffee Development Authority (UCDA) standards define grades of qualities in the limited set: A B C, and D.

02 MILESTONE TWO

System Design & Entity-Relationship Modelling

3.1 Database Design Approach

Before drafting a single SQL statement, the team dedicated a large amount of time to system level planning. The very first output was an Enhanced Entity-Relationship Diagram (EERD) that describes the whole set of entities, their properties associations cardinality restrictions, and any specialization hierarchies being part of the domain. The EERD was created with draw.io and was used as the main source of truth for the entire development process. Each table, foreign key, and constraint in the final MySQL schema can be directly linked to a purposeful design decision in the EERD.

3.2 Key Design Decisions

PERSON as a Shared Supertype

FARMER and EXTENSION_WORKER are two categories of PERSON. If you think in terms of database tables, one single row in the PERSON table (that would be the one with the key national_id) can be connected to one single or one multiple rows in the two other tables (FARMER and EXTENSION_WORKER). This will not only avoid the duplication of the same information but, a single NIN will also become the unique identity anchor across the entire system.

Disjoint Specialisation for Extension Workers

An extension worker can either be a FIELD_OFFICER or a TRAINER - these two roles are mutually exclusive. The disjoint constraint is implemented through separate subtype tables rather than using a single role column, which allows each specialization to have its own separate set of attributes.

Plot-Level Production Tracking

Production data isn't recorded on the FARM level but at the FARM_PLOT level, which allows very detailed tracking different coffee varieties grown on parts of a single farm. With this, the design enables the quality grade system by UCDA (A B C, D) to be implemented at the most detailed level possible.

Geographic Hierarchy

Location data is structured as a three-level hierarchy: REGION → DISTRICT → SUBCOUNTY. This mirrors Uganda's administrative structure and allows the system to aggregate data at any geographic level for reporting and extension worker zone assignment.

3.3 Entity Inventory

The EERD captures the following core entities and their key attributes:

Entity	Key Attributes
PERSON	`national_id (PK), full_name, gender, dob, phone, email`
FARMER	`farmer_id (PK), national_id (FK), land_size_acres, subcounty_id (FK), cooperative_id (FK)`

Entity	Key Attributes
EXTENSION_WORKER	worker_id (PK), national_id (FK), qualification, zone_id (FK)
FIELD_OFFICER	Subtype of EXTENSION_WORKER – linked to FARM_VISIT records
TRAINER	Subtype of EXTENSION_WORKER – linked to TRAINING_PROGRAMME records
FARM	farm_id (PK), farmer_id (FK), total_area_acre, gps_coordinates
FARM_PLOT	plot_id (PK), farm_id (FK), coffee_variety, area_acres, planting_year, num_trees
PRODUCTION_REPORT	report_id (PK), plot_id (FK), harvest_season, quantity_kg, quality_grade
FARM_VISIT	visit_id (PK), worker_id (FK), farmer_id (FK), visit_date
VISIT_REPORT	report_id (PK), visit_id (FK), observations, recommendations
TRAINING_PROGRAMME	programme_id (PK), trainer_id (FK), topic, venue, start_date, end_date
PROGRAMME_ENROLLMENT	enrolment_id (PK), programme_id (FK), farmer_id (FK), completion_status
PRODUCT	product_id (PK), product_name, unit, stock_quantity
SUPPLIER	supplier_id (PK), name, location, certification
DISTRIBUTION	dist_id (PK), visit_id (FK), product_id (FK), quantity, dist_date
AUDIT_LOG	log_id (PK), action_type, table_name, record_id, changed_by, timestamp

03 MILESTONE THREE

Database Development & Implementation

4.1 Implementation Environment

The database was implemented using MySQL, running locally on each group member's development machine. The complete SQL script is designed to be executed in full from the MySQL command line or from within MySQL Workbench. The following tools were used throughout development:

- **MySQL 8.0** — Primary relational database management system for all schema, views, triggers, and procedures.
- **MySQL Workbench** — Schema design, ER diagram generation, and interactive query testing.
- **draw.io** — Construction and iteration of the Enhanced Entity-Relationship Diagram.
- **Visual Studio Code** — SQL script editing with syntax highlighting and version control integration.

4.2 Database Views

Three database views were created to simplify the most common reporting queries and to provide a clean abstraction layer over the complex multi-table joins that underlie the schema. Views also serve a security function,

View 1 — vw_farmer_profile

Aggregates farmer identity, location, land size, and cooperative membership into a single queryable result set. This view is the primary entry point for all farmer-facing reports.

```
CREATE VIEW vw_farmer_profile AS
SELECT
  f.farmer_id,
  p.full_name,
  p.gender,
  p.phone,
  p.email,
  f.land_size_acres,
  s.subcounty_name,
  d.district_name,
  r.region_name,
  c.coop_name
FROM FARMER f
  JOIN PERSON     p ON f.national_id   = p.national_id
  JOIN SUBCOUNTY  s ON f.subcounty_id  = s.subcounty_id
  JOIN DISTRICT   d ON s.district_id   = d.district_id
  JOIN REGION     r ON d.region_id     = r.region_id
  LEFT JOIN COOPERATIVE c
    ON f.cooperative_id = c.cooperative_id;
```

View 2 — vw_production_summary

Joins FARM_PLOT and PRODUCTION_REPORT with farmer identity to provide a consolidated production dashboard. Used to generate Yield Forecast reports for the Ministry.

```
CREATE VIEW vw_production_summary AS
SELECT
  p.full_name,
  fp.coffee_variety,
  pr.harvest_season,
  SUM(pr.quantity_kg)  AS total_kg,
  pr.quality_grade
FROM PRODUCTION_REPORT pr
  JOIN FARM_PLOT fp ON pr.plot_id    = fp.plot_id
  JOIN FARM      f  ON fp.farm_id    = f.farm_id
  JOIN FARMER    fa ON f.farmer_id   = fa.farmer_id
  JOIN PERSON    p  ON fa.national_id = p.national_id
GROUP BY p.full_name, fp.coffee_variety,
         pr.harvest_season, pr.quality_grade;
```

View 3 — vw_input_distribution

Provides a monthly Support Distribution report linking each farmer to the inputs they have received, the visit through which they were distributed, and the responsible extension worker.

```
CREATE VIEW vw_input_distribution AS
SELECT
  p.full_name        AS farmer_name,
  prod.product_name,
  d.quantity,
  d.dist_date,
  pw.full_name       AS officer_name
FROM DISTRIBUTION d
  JOIN FARM_VISIT    fv  ON d.visit_id    = fv.visit_id
  JOIN FARMER        fa  ON fv.farmer_id  = fa.farmer_id
  JOIN PERSON        p   ON fa.national_id = p.national_id
  JOIN PRODUCT       prod ON d.product_id  = prod.product_id
  JOIN EXTENSION_WORKER ew ON fv.worker_id = ew.worker_id
  JOIN PERSON        pw  ON ew.national_id = pw.national_id;
```

5.1 Overview

Security and automation are implemented at three complementary levels in the Coffee Management System.

04 **MILESTONE FOUR**

Security, Automation & Role-Based Access Control

Together, they ensure that data entering the database is valid, that all sensitive operations are traceable, and that each user can only do what their role requires nothing more.

5.2 Stored Procedures

Stored procedures group SQL statements and business logic into single, reusable units that execute inside the database engine. They serve as the only sanctioned entry points for data modification, ensuring consistency regardless of which application or user initiates an operation.

sp_register_farmer — Farmer Registration

Atomically inserts a PERSON row and a FARMER row inside a single transaction. If either insert fails, the entire operation is rolled back and the error is re-raised to the calling application.

Additional Stored Procedures

The following additional procedures complete the automation layer:

Procedure	Responsibility
`sp_record_farm_visit`	Validates that the field officer is assigned to the correct geographic zone before creating a FARM_VISIT record. Returns the new visit_id.
`sp_record_production`	Checks that the referenced plot exists and that the quality grade is within the permitted UCDA range (A–D) before inserting a PRODUCTION_REPORT row.
`sp_distribute_product`	Checks current stock level, reduces inventory via a row-level lock, and records the distribution event. Raises an error if stock is insufficient.
`sp_enroll_farmer`	Registers a farmer in a training programme after verifying that the programme has not yet reached maximum capacity.

5.3 Triggers

Triggers fire automatically in response to INSERT, UPDATE, or DELETE events. They enforce business rules and referential constraints that go beyond what SQL constraint syntax alone can express.

trg_plot_area_check — Plot Area Validation

Fires BEFORE INSERT on FARM_PLOT. Sums the existing plot areas for the farm and compares the total against the farm's declared total acreage. If the new plot would cause the total to exceed the farm's declared area, the insert is blocked.

```
CREATE TRIGGER trg_plot_area_check
BEFORE INSERT ON FARM_PLOT
FOR EACH ROW
BEGIN
  DECLARE v_total_area  DECIMAL(10,2);
```

```
  DECLARE v_used_area   DECIMAL(10,2);

  SELECT total_area_acre INTO v_total_area
  FROM   FARM WHERE farm_id = NEW.farm_id;

  SELECT IFNULL(SUM(area_acres), 0) INTO v_used_area
  FROM   FARM_PLOT WHERE farm_id = NEW.farm_id;

  IF (v_used_area + NEW.area_acres) > v_total_area THEN
    SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT =
        'Plot area exceeds available farm area.';
  END IF;
END;
```

5.4 Role-Based Access Control

Five database user accounts were created, each mapped to a distinct operational role. Every account is granted only the privileges strictly necessary for its function, following the Principle of Least Privilege.

Role / Account	Privileges Granted
coffee_admin	ALL PRIVILEGES on coffee_mgmt.*. Used exclusively by the DBA for schema maintenance. Never used in day-to-day operations.
Extension_worker	SELECT on all tables. INSERT and UPDATE on FARM_VISIT, VISIT_REPORT, PRODUCTION_REPORT, VISIT_DISTRIBUTION. EXECUTE on sp_record_farm_visit and sp_record_production.
farmer	SELECT on all tables and all views. No INSERT, UPDATE, or DELETE access whatsoever. Used for reporting and data export queries only.

6.1 Summary

The Coffee Management System represents a complete, production-ready relational database designed for the administrative and operational needs of a national coffee agricultural support programme. The system was built from first principles beginning with a rigorous Enhanced Entity-Relationship Diagram that captured specialisation hierarchies and cardinality constraints, then translated into a fully normalised MySQL schema, secured with role-based user accounts, and automated with stored procedures and triggers.

Every design decision from the choice of PERSON as a shared supertype, to the row-level lock in sp_distribute_product, to the plot area trigger was made with real-world operational requirements in mind. The audit log ensures accountability. The stored procedures ensure consistency. The privilege model ensures that each user sees and does only what their role requires.

6.2 Achievements by Milestone

Milestone 1 — Requirements

- ✓ Six functional requirement categories fully documented and agreed.
- ✓ Nine data entity groups identified and scoped for implementation.
- ✓ Ten explicit assumptions documented to bound system scope.

Milestone 2 — System Design

- ✓ EERD constructed with 15+ entities, covering all functional areas.
- ✓ Specialisation hierarchies (PERSON → FARMER / EXTENSION_WORKER) correctly modelled.
- ✓ Three-level geographic hierarchy implemented to reflect Uganda's administrative structure.

Milestone 3 — Database Development

- ✓ Full MySQL schema implemented and tested in MySQL Workbench.
- ✓ Three views created: vw_farmer_profile, vw_production_summary, vw_input_distribution.
- ✓ All joins, foreign keys, and normalisation rules verified against the EERD.

Milestone 4 — Security & Automation

- ✓ Five stored procedures implemented with full transaction management and error handling.
- ✓ Three triggers enforce business rules at the row level automatically.
- ✓ Five role-limited database user accounts implement least-privilege access control.
- ✓ Audit log automatically records all sensitive data changes with user and timestamp.

The system is ready to receive real data. The Coffee Management Database delivers a secure, consistent, and auditable platform that can be deployed immediately to support Uganda's national coffee agricultural services programme.

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